FLEET TEAM DIGEST

737MAX-FTD-34-23005

Issue Title : 737MAX with FMC U14+ "Future Time" downlink issues while using ATN PM CPDLC in Eurocontrol regions

Airplane Model	ATA	Minor Model(s)
737MAX	3461-00	-10, -7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
10/12/2023	09/18/2026	11/14/2024
Estimated Completion Date	**Next Update Date**	
03/29/2028	12/19/2026	

Revision Description

Updated Milestone for MAX FMC U14.2 release to 1Q2028

Applicability

Applies to 737 MAX airplanes.

Description

A 737MAX operator with Collins PM-CPDLC (Protected Mode – Controller Pilot Datalink Communication) CMU configuration has reported "Future Time" issues while using ATN (Aeronautical Telecommunications Network) CPDLC over EUROCONTROL regions. Per EUROCAE ED-110B ATN Baseline 1 requirement.

- When the airborne system receives a CPDLC message with timestamp that indicates a future time greater than 2 seconds from the current time, it shall send a CPDLC downlink message containing the concatenation of message element DM62 (ERROR (error information)) with the choice (2) followed by message element DM98 (UPLINK TIMESTAMP INDICATES FUTURE TIME).

While ED-110B specifies no more than a 2-second time deviation between CPDLC message timestamp and airborne system time, in practice, ATN avionics suppliers have chosen to implement a 3-second compliance time. For 737NG/MAX, two ATN CPDLC configuration options are offered: CMU-based PM-CPDLC and FMC-based FANS-2 (FMC U14 or later).

In the CMU-based PM-CPDLC implementation, the CMU hosts the ATN application and uses FMC time to determine ED-110B compliance. In the FMC-based FANS-2, the FMC hosts the ATN application which determines ED-110B compliance with FMC time. The FANS-2 capable CMU is in ‘router’ configuration and does not check for deviation between the message timestamp and the system time.

In the reported cases, the FMC UTC time lags by approximately 2-4 seconds. When the time requirement is not met, the CMU downlinks "Uplink Timestamp indicates Future Time" to the ATC (Air Traffic Control) Center. In the 737MAX configuration (only) with FMC U14 or later, the FMC Operational Program Configuration (OPC) provides the "Autonomous GPS Select" software option which can be enabled or disabled. When the "Autonomous GPS Select" option is disabled, the FMC provides the "Hybrid GPS"

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feature. When the "Autonomous GPS Select" option is enabled the FMC provides the "Direct GPS" feature, which is the 737NG/ MAX baseline option.

The "Hybrid GPS" feature was introduced in FMC U14 for 737MAX (only) and this feature uses the hybrid GPIRU (GPS/Inertial Reference Unit) output from the ADIRU (Air Data Inertial Reference Unit) in the navigation solution - refer to the Ref a) and Ref b) service letters for more information. When the "Direct GPS" function is enabled, the FMC does not use the hybrid GPIRU output in the navigation solution. Rather, the FMC sources GPS input directly from the MMR GNSS output. The FMC UTC time and date are sourced from the MMR GNSS output when in Direct GPS configuration and the ADIRU GPIRU output when in Hybrid GPS configuration. The Ref b) service letter describes details of the Hybrid GPS and ADIRS feature and activation.

Two CMU configurations are available with the 737MAX ATN CPDLC datalink configurations:

1- Collins Hardware Part Number 822-1239-151

2- Honeywell Part Number 965-0758-006

FANS-1 (CPDLC, ADS-C) does not include this CMU time requirement. In the FANS-1 configuration, the ATN datalink function is not activated therefore FANS-1 configuration is not affected.

Status

Systems Integration testing was conducted at Boeing labs and found that the Hybrid GPS implementation introduces additional propagation delay due to the added integration of the ADIRU for the Hybrid GPIRU output. Supplier specific design philosophy is also a contributing factor to differing behavior across configurations.

The following behavior was exhibited in the Boeing Labs:

1-When the 737 MAX FMC OPC is configured for Direct GPS, the FMC UTC time is delayed approximately 1 second.

2. When the 737 MAX FMC OPC is configured for Hybrid GPS, the FMC UTC time is delayed 2-4 seconds.

Lab Test Conclusions & Observations (See Lab Testing Results attachment):

-Collins PM-CPDLC CMU is most susceptible to the ‘Future Time’ issue.

-FMC FANS-2 and Honeywell PM-CPDLC CMU do not exhibit the ‘Future Time’ issue unless a manually inducing 1-2 second delay is introduced.

- Based on real-world variabilities, such as ATN network latency, the FMC FANS-2 and Honeywell PM-CPDLC CMU configurations could experience additional time delay similar to the additional manually induced delays from the lab testing.
- FMC FANS-2 and Honeywell PM-CPDLC CMU could encounter the ‘Future Time’ issue in the field, but it is expected to be significantly less frequent than Collins PM-CPDLC CMU.

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Interim Action

If in Collins PM-CPDLC CMU and FMC OPC enabled 'Hybrid GPS' configuration ('Autonomous GPS Select' option disabled) and operating in ATN airspace, Boeing recommends operators to submit a service request via the Boeing Communication System (BCS) or Field Service Representative (FSR) to request a no-charge Service Bulletin to revert to the Direct GPS ('Autonomous GPS Select' option enabled) configuration.

If in Honeywell PM-CPDLC CMU or FMC FANS-2 with FMC OPC enabled 'Hybrid' GPS configuration ('Autonomous GPS Select' option disabled) and operating in ATN airspace, Boeing offers a choice to retain the Hybrid GPS configuration or revert to the Direct GPS ('Autonomous GPS Select' option enabled) configuration. Operators may submit a service request via the Boeing Communication System (BCS) or Field Service Representative (FSR) to request a no-charge Service Bulletin to revert to the Direct GPS ('Autonomous GPS Select' option enabled) configuration at discretion.

If not operating in ATN Airspace or not configured for ATN CPDLC Datalink, no interim action is needed.

Final Action

A design improvement will be implemented in FMC OPS U14.2.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.
Interim Action: October 2023 (Complete)
Root Cause Established: December 2023 (Complete)
Solution Selected: JAN 2024 (Complete)
MAX Service Bulletin Available: 1Q2028

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Letter (SL)	737-SL-34-251		
Service Letter (SL)	737-SL-34-263		

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Related Categories

- Flight Deck Effects and Maintenance Messages : Issue that occurs frequently and causes a flight deck effect or maintenance message

Attachments:

a) Test Configurations

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